RESPONSE TO REVIEWERS

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Revised title: *Two Reachability-Gated Relations in MST-wi: Temporal Mismatch, Oracle Admissibility, and Measurement Limits*

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Dear Associate Editor and Reviewers,

Thank you for the detailed reports. The revision is substantive rather than cosmetic, and we have kept the narrowing that both reports asked for: the title, the abstract, the running head and every claim are now restricted to the two reachability-gated relations that were actually examined. Three statements that were previously argued in prose are now computed from records and reproduced by script—the complete factorial design and its reconciliation, the truth table of the two relations, and a competitive differential baseline on a matched request budget. One experiment that had not been run at all (a competing differential baseline) has now been run and reported. Where a request could not be met with the evidence we have, we say so in the manuscript instead of softening the text; eight of the twenty-one items are met only in part, and Section **??** of this letter lists them together with what is missing and why.

References below are to the revised manuscript (26 pages, 22 tables). Table and section numbers are stable; page numbers are given for the tables most often at issue. Every number quoted in this letter is generated by a script from the raw records; none is typed by hand, and the generation scripts are part of the reproducibility package.

1 Reviewer 1

1.1 Novelty positioning against existing BOLA testing research

Concern. The paper identifies limitations in reachability-based adjudication, oracle assumptions and metric design, but does not sufficiently distinguish these contributions from established work on oracle problems, stateful web testing, authorization testing and metamorphic testing. A structured comparison is requested.

Response. We agree that "a frozen snapshot goes stale" is not new, and the manuscript now says so explicitly. What we can support with evidence is narrower and is stated as three falsifiable increments rather than as a general positioning claim. (i) *Predicate level:* at source, the verdict of each of the two relations is a function of the crawl-time reachability model and the request outcome, so it is determined by quantities that a direct request-plus-oracle comparison already has (Section VII-B, confirmed by execution in Section VII-C). (ii) *Subset level:* the false proofs we exhibit are confined to the cells whose grant is invisible to the framework's own record, and the channel decomposition (Table 11, p. 13) shows exactly which channels those are; neither product's membership surface produces a single false *allow*. (iii) *Denominator level:* a pooled input-level specificity cannot resolve a failure confined to one class, and no fixed threshold can force such a class below it (Proposition 1 and Corollary 1, Section IV-E). We also state plainly that we do not compare detection rates and cannot, from our evidence, claim any.

Action. Section IX rewritten around these three increments; the claim of being "orthogonal" to prior work deleted; the abstract's first sentence and the conclusion restated in the same terms.

1.2 A complete definition of the false-proof outcomes

Concern. The mapping between alarms, false alarms and proven violations and the ground-truth authorization state is not formalized consistently; decision variables and truth conditions are requested for every combination of request outcome, actual entitlement, oracle response and detector verdict, together with an account of how a legitimate successful request is distinguished from unauthorized access without letting the tested reachability model determine its own ground truth.

Response. Table 16 (Section VI-B, p. 17) is now that table. It defines four quantities over the 712 temporal cells: Y (the ordinary request succeeds), M (the crawl-time model lists the object for the subject), O (the introspection surface's answer, with an explicit "no definite answer" value), A (the alarm of either relation; on this domain both reduce to $y \wedge \neg m$) and D (the direct adjudicator, $y \wedge \neg o$), plus $H = A \wedge \neg o$, the promoted verdict of the state-calibrated method, which exists only where O is definite. The twelve rows cover every combination of (Y, M, O), and the count columns split each row by the ledger's entitlement E, so each row states its truth conditions for both values of E. On the anti-circularity point: the label E comes from the framework's own grant and membership ledger, which is an independent

entitlement source; M is an *input* to the relations and is never used as a label. A legitimate success is separated from unauthorized access only by an independent *deny* from E together with a content-level *allow* observation—never by M—and the manuscript says that the relation's alarm is not evidence of illegal access.

Action. Table 16 and the definitions in Section IV-A; the three-way split of the native run into A (the engine's own alarm), H (our promotion policy) and T (confirmed) is given in Table 21 (p. 21).

1.3 Characterization of authorization-state transitions

Concern. The three-phase protocol does not comprehensively model revocation, inherited permissions, public visibility, role changes and concurrent updates; the state-transition semantics, the timing of each observation and the consistency assumptions behind the deterministic false proofs are requested, as is whether the conclusions hold when authorization changes between request and adjudication.

Response. Section V-A now states the semantics that the experiment actually exercises, and three scope limits are stated as part of the result rather than buried in the discussion. The eight committed transitions are a designed set (direct grant and revoke, team attach and detach, member addition and removal, private-to-public visibility, site-administrator grant and revoke); each cell is serialised, no mutation is issued between a cell's request and its probe, so the window between them is static *by construction*; and a deployment in which a grant or revoke lands inside that window is outside what these cells measure—for such a case the cell's conclusion does not apply and the observation must be repeated, which the manuscript says we do not claim to detect. Revocation and inheritance behave differently from grants in this design and the table shows it: the revoke rows produce no stale false proof at all (Table 12, p. 14), because an over-listing model makes the relation silent (a miss) rather than wrong (a false proof). On concurrency, the harness is single-threaded, no concurrent writer ran, and *hv* is declared to be a harness-maintained logical version rather than an atomic server snapshot—which is what makes the still window meaningful.

Action. Section V-A (Table 12, p. 14); the same scope limits restated in Section VIII.

1.4 Operationalizing the admissibility condition

Concern. Clause (2)—that the oracle is not affected by the same defect under investigation—remains a qualitative judgement; a reproducible procedure for determining admissibility *before* examining outcomes is requested.

Response. We agree this is the weakest link, and we have not solved it. What we did do: the candidate surfaces, the granting channels and the defect classes were fixed in a protocol file whose digest is bound into the run record before the reported run; the primary-vantage gate was checked (4 of 4) before that run; and the one amendment made afterwards is bound by digest to the run record, with the contradiction in its own timestamp disclosed rather than backdated—the amendment file's `written_at` postdates the last probe, and the manuscript says so and gives the commands to re-check it. What we did *not* do is supply a mechanical, implementation-independent test of independence from the tested mechanism. Section VI-E now says explicitly that clause (2) is a per-system judgement and is not automatable from readability alone, and it makes the judgement concrete instead of abstract: Gogs and GitLab fail it, and the failure is observable as false proofs on identifiable channels (Tables 11 and 13). We would rather state the limit than imply a procedure we do not have.

Action. Section VI-E; the pre-registered protocol and the amendment timing note are in the reproducibility package.

What remains. No mechanical independence criterion. This is listed as a limitation in Section VIII.

1.5 Proposition 2: separating the definitional from the empirical

Concern. The proposition risks being tautological; the formal identity should be distinguished from the empirical premise that the evaluated relations consume no additional decision-relevant information, with a truth-table derivation or explicit functional representation.

Response. The result is now split in two. **Lemma 1** (absorption) is the algebraic part: if a verdict is a function of an oracle answer and an ordinary request, it cannot outperform the direct consumer of those same inputs. That part is definitional and is labelled as such. **Proposition 2** is the empirical part: the two relations fall inside the lemma's domain, which is a claim about this implementation and is checked, not assumed. The explicit predicates arewritten out ($A_{002} = \neg sup(B,A) \wedge \neg m \wedge \neg err(o_1) \wedge q(o_1,o_2)$ and $A_{004} = y \wedge \neg m$, against $D = y \wedge \neg o$), and on the domain studied both reduce to $y \wedge \neg m$. The reduction was verified mechanically against 576 constructed records and 712 temporal cells with zero mismatches; the generating script exits rather than emitting the table if any record disagrees. The equivalence is stated correctly as $A = D \Leftrightarrow y = 0 \vee m = o$: where D is defined the two verdicts coincide in 530 of the 584 readable cells and part in the two rows with Y=allow and $M \neq O$—40 cells where the model is right and the oracle

wrong, and 14 where the model is stale and the oracle right. Those two rows are the two failure mechanisms the cross-vendor section keeps apart. The falsification condition is stated: produce an observable, outside what a succeeding request plus a readable oracle can determine, on which an authorization metamorphic relation still decides.

Action. Section VI-B with Lemma 1, Proposition 2, Table 16 (p. 17) and the corrected equivalence; the earlier claim of extensional equality is deleted.

1.6 Coverage and identifiability of the 576-record enumeration

Concern. A transparent mapping from configurations and scenarios to all records is requested, with the complete factorial design, an account of how each factor is varied, attribution of the reported effects, and an explicit reconciliation of the 36/18 representative grid with the later 312/264 partition.

Response. The enumeration is now inspectable end to end. Table 1 (p. 4) is the complete design: seven factors, their levels, how each level is varied, the record count per level and what it crosses with, with the additive identity $44 \cdot 2 \cdot 3 \cdot 2 + 8 \cdot 3 \cdot 2 = 576$ in the footnote. Table 19 (p. 20) is the configuration-by-scenario census, where a dash marks an inadmissible combination and the caption says why (a configuration with no share has nothing to list; a listed cell requires introspection on). The reconciliation the reviewer asks for is Table 5 (pp. 8–9): it starts from the S3–S7 original grid of 54 records (which contains 8 v2 false proofs) and adds the remaining blocks, asserting by record key that the blocks are mutually disjoint and that their union is exactly 576, split 312 readable / 264 unreadable. Section III-A states what the design can and cannot identify: the three dimensions are nested rather than crossed (the relation is single-valued on S5; channel and visibility are defined only inside one exception setting), so a full interaction estimate is not available—which is why attribution is done with matched pairs (Table 2, p. 5) rather than with a regression.

Action. Tables 1, 2, 5 and 19; Section III-A.

1.7 Comparison against competitive black-box methods

Concern. `direct_only` is deliberately unfair and `obs_success` and `always_alarm` are intentionally weak; a differential, multi-account authorization tester probing equivalent objects under distinct privilege levels is requested, with matched query budgets, execution costs and detection outcomes.

Response. This is the one comment that required a new experiment, and it has been run. Section VI-D and Table 17 (p. 19) report three conventional differential rules on the same 42 controlled cells (six configurations crossed with S1–S7) and, critically, on the *same probe pair*: every method consumes one request as the owner and one as the actor; ours issues one additional read of the authorization surface on the 27 cells where a relation alarms and none where it does not (0.64 per cell on average). *B1 status-only* alarms whenever the replayed request returns a non-error. *B2 equality replay* alarms only when the owner succeeded and the actor received byte-identical output—the differential rule with no authorization model. *B3* adds the strongest information a tester can hold without reading the target: it suppresses every cell covered by the tester's own share log. On these cells B1 and B2 claim on all 27 allowed cells with 12 false proofs each (precision 55.56%), B3 claims 21 with 6 false proofs (71.43%), all six on the out-of-band channel, and ours claims 10 with no false proof but 17 abstentions (coverage 59.52%). Two readings are explicitly ruled out in the text: this is *not* a detection gain, because the baselines find all fifteen violations while ours confirms the ten whose surface is readable and declines the other five; and it is *not* a free improvement, because coverage falls to 59.52%. A corroborating observation on a real system is reported with the same qualification: on the Gitea snapshot the equality rule fires on all 25 legitimately allowed cells and the in-band log still leaves 24, because the framework log does not observe 0.96 of the effective access there—a property of the constructed topology, not an estimate of field prevalence.

Action. Section VI-D and Table 17 (p. 19); the runner (`experiments/run_baseline.py`) and its raw output are in the reproducibility package, and both baseline tables are reproduced byte-identically by `verify_artifact.sh repro`.

What remains. Execution cost is reported as request counts, not wall-clock time, for these four methods: all of them replay the same recorded tape, so a timing comparison would measure our replay loop. End-to-end timing is reported only for the real-system temporal loops (Gitea 21.2 s, GitLab 7.3 s, of which the probes are 20.7 s and 6.0 s).

1.8 Channel-by-channel isolation of the oracle-validity problem

Concern. A channel-by-channel confusion matrix separating direct membership, team-derived access, administrative grants, public visibility and out-of-band sharing is requested, together with a distinction between an incorrect oracle response and a stale reachability model.

Response. Table 11 (Section V-B, p. 13) is that matrix: the owner-vantage introspection surface scored against the ledger per channel and per product, with cells, agreements in each direction, the two error directions, and a separate "unanswered" column for anonymous readers. Neither product produces a single false *allow*: the membership surface errs only by omission, and public and site-administrator cells are exactly the channels that are not group or repository membership. The two failure mechanisms are kept apart by construction and by table: in the truth table, row 6 is the oracle answering *deny* on a cell the ledger allows (40 cells, an unsound oracle), row 7 is the model failing to list a cell the ledger allows (14 cells, a stale model), and the two verdicts disagree in precisely those rows. On Gogs: the identical three-phase protocol was run (Table 13, p. 14), but Gogs' introspection surface exposes no per-channel decomposition equivalent to the other two products, so the manuscript declares the gap instead of manufacturing the matrix.

Action. Tables 11 and 13; the gap stated in Section VIII.

What remains. No per-channel four-cell matrix for Gogs.

1.9 Detection results across independently implemented defects

Concern. The three seeded instances share one fault pattern; the positive control should be expanded to multiple independently implemented authorization defects, including unauthorized reads, unauthorized modifications, inherited-permission errors and privilege escalation, with per-defect recall and false-positive rates.

Response. We did not implement multiple independent defects, and the manuscript says so. What was added is a real vulnerable/fixed pair rather than another seeded instance: a publicly fixed defect in Gitea (object-versus-path ownership in the comment-permission check, fixed between 1.20.5 and 1.20.6), driven through the same adjudication gate. The verdict uses the ledger's entitlement and the content-level observation: on 1.20.5 the gate yields one true positive and no false proof; on 1.20.6 it yields no alarm and no claim; the twelve control checks (ten expected responses across the two builds and two recovery checks) all behave as expected, and the generating script refuses to emit a number if any control deviates. The manuscript is explicit that this is a positive control of the *gate*—an independent entitlement deny together with a content-level allow—and not evidence about the relations' recall on real defects, and the conclusion states that a stronger claim needs a *population* of real defects, which a single pair cannot provide. The seeded control is also bounded honestly: the three instances instantiate one fault pattern, and the one BFLA defect we seeded (a non-privileged actor succeeding at a collaborator-write endpoint) is reported as a boundary of the read-oriented logic, not as a detection failure.

Action. Section V-D and Table 15 (p. 15); Section V-C and Table 14 (p. 14) for the seeded control; the scope statement in Section X.

What remains. Per-defect recall and false-positive rates across several independent defect classes. Not done, and stated as out of scope in the manuscript.

1.10 Control of temporal and observational effects

Concern. The ordering of crawling, entitlement mutation, oracle querying and request execution must be specified, with timestamps or event logs sufficient to verify that the comparison changes only the model-reading time, and an account of how live-state queries avoid changing the tested state or gaining privileges.

Response. The ordering is fixed and stated: crawl, then freeze the model at t_0, then commit the transition, then query the oracle, then issue the request; all observations are `GET` requests, so nothing in the measurement path mutates state; the recrawl uses the same credential as the original crawl, so it grants no privilege the detector did not have; and the owner oracle is reported separately from the subject-side reading rather than pooled with it. The run record carries the protocol digest, the event sequence numbers, the policy version, the M_0/M_1 model digests and the per-request timestamps, and the harness version *hv* is declared to be a logical version rather than an atomic server snapshot. The results are symmetric across the two products: Gitea 280 cells with 13 stale false proofs, all 13 repaired by a recrawl with the same credential and 0 mismatches under the live reading; GitLab CE 432 cells with 16 stale false proofs, all 16 repaired and 0 under live reading.

Action. Section V-A and Table 12 (p. 14); the pre-registered protocol file, the run records and the amendment timing note are in the reproducibility package.

1.11 The specificity denominator

Concern. The 84.62% figure assumes two follow-up inputs per case and cannot establish the numerical behaviour of the published 99.81% figure; the original counts or a parameterized sensitivity analysis are requested, with a clear

distinction from conventional specificity.

Response. The published figure is now treated as what it is: a specificity, $\mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$, over the follow-up inputs F, with the manual verification of the reported failures acknowledged—so the labels are not the crawler's own answer, and we do not attribute the number to a circular label. The 84.62% illustration has been removed from the manuscript entirely, together with the claim that it agrees with the crawler. We cannot recover the engine's exact input count, so instead of substituting a value we parameterize: with $\mathrm{FN}=0$ the figure equals $1-\mathrm{FP}/(F-\mathrm{TP})$, a class supplying a share w of the negative inputs can move it only within $[1-w,1]$, and no fixed threshold $t<1$ can guarantee that a class-uniform failure pushes it below t (Proposition 1 and Corollary 1, Section IV-E). The proposed scenario-level quantity is labelled an analytical construct for characterising resolution, not a replacement metric, and the text says the finding is about what the published figure can be asked to show, not about its arithmetic.

Action. Section IV-E; restated in Section VI-C; the figure removed from the body.

1.12 Heterogeneous scenario weights

Concern. Proposition 1 assumes equal contributions per scenario; the result should be extended to unequal request counts, correlated inputs and different alarm counts per scenario, including the case where the number of generated inputs depends on scenario complexity or detector behaviour.

Response. The fixed-weight case is derived (Proposition 1 and its corollary, Section IV-E) and the arbitrary-share case is bounded without any equal-weight assumption: whatever the class shares are, a failing class contributing a share w of the negative inputs moves the figure only within $[1-w,1]$ (Section VI-C). We did *not* derive the endogenous case in which the number of generated inputs is itself a function of detector behaviour, because our records are a fixed enumeration in which the input count does not respond to the detector; deriving it would require a pipeline whose generation is detector-dependent, which we did not build. Rather than assert a generality we cannot support, Section VIII states the limitation: records within a design cell are not independent, the records are a fixed enumeration, and we therefore report counts and rates rather than inferential intervals, with no result resting on statistical significance.

Action. Section IV-E and Section VI-C; the limitation stated in Section VIII.

What remains. No derivation for detector-endogenous input counts.

1.13 Consistent treatment of abstentions and operational cost

Concern. Perfect precision bought by abstention is not characterized by precision and recall alone; coverage, selective risk, abstention rate and performance conditional on oracle availability are requested, alongside execution time and request overhead, with a distinction between a safe abstention and a successful detection.

Response. Table 4 (Section IV-D, p. 8) rescored the same 576 records on the *decision* rather than on the alarm, and reports coverage, the abstention count and rate split by whether the record held a true violation, false proofs, misses and selective risk, with the explicit warning that risk is comparable across methods only at equal coverage—which these methods do not have. The state-calibrated method covers 60.42% and abstains 228 times; panel (b) splits those abstentions by reason, and only the first row (156 surface-unreadable, 108 of them true violations) is the availability gate the method is named for, the other 72 being a false proof converted into a non-decision. The direct methods abstain under the same unreadable-means-abstain contract (coverage 54.17%, 264 abstentions), and the forced-negative reference—coverage 100%, 108 misses, risk 18.75%—is labelled as the price of forcing a verdict on unreadable cells, not as a property of any method. A passed record is defined as one on which the method did not fire, explicitly not a certificate of safety, and the text refuses to rank by risk alone, noting that the always-degrade reference has the second-lowest risk while detecting nothing. Cost is reported as request counts: 4833 enumeration requests (median 8, range 1–17) and 104 probe `GET`s (52 cells $\times$ 2).

Action. Table 4 (p. 8) and Section IV-D; the abstention semantics unified across methods in Section IV-A.

What remains. Wall-clock execution time for the replay-based methods, for the reason given under comment 7.

1.14 Reproducibility of the principal claims

Concern. A version-pinned environment, exact commands, complete configuration and seed files, machine-readable raw outputs and scripts that regenerate every table and figure are requested, together with documentation of the two failed relation executions, the engine-side defect and the treatment of missing or failed observations.

Response. The package now carries a pinned environment, the exact commands, the seed and configuration files, the machine-readable raw records and the scripts that regenerate every table and figure, and it was verified from a clean

copy rather than asserted: in an empty directory the submission package compiles (26 pages, no undefined references), and `verify_artifact.sh repro` reproduces all 28 generated artefacts byte-identically, including the two baseline tables. The native run is reported as three stages rather than one number (Section VII-C, Tables 20–22, pp. 21–22): A, the engine's own `getFailures()` alarms, 6; H, our promotion policy applied to them, 6; T, confirmed, 0—with 6 false proofs. The content predicate is reported by invocation count rather than claimed clean, and the text says so. The two failed relation executions and the engine-side defect are documented at source, and the treatment of missing or failed observations is written into the protocol file that the run record binds.

Action. Section VII-C and Tables 20–22; `reproduce.sh`, `verify_artifact.sh` and the hash manifests in the package.

What remains. The failure and missing-observation rules live in the protocol file and the scripts; Section VII states them only in part.

1.15 Narrowing the conclusions; an evidence-to-claim table

Concern. Statements suggesting that adjudication of genuine violations is absent from the evaluated relations risk extending beyond the experiments; conclusions established by source-level dependence, by the synthetic construction and on real systems should be distinguished, with a final table linking each research question to its assumptions, tested configurations, results and limitations.

Response. Table 18 (Section VI-E, p. 20) is that table. Each row names the assumption the claim needs, the artefact that carries it, and the boundary that remains; the last row is the adjudication gate and is marked as not being evidence about detection effectiveness. The three findings are explicitly attributed to different kinds of evidence—source-level functional dependence, a constructed enumeration, and two real-system studies—and the text says a reader should be able to see which kind carries which sentence. The sentence that extended beyond the experiments has been deleted, together with the mis-attribution of a constructed result to a published one; the title, abstract, running head and conclusion are narrowed to the two reachability-gated relations, and each finding is stated with its own falsification condition.

Action. Table 18 (p. 20); Section VI-E; title, abstract and Section X.

2 Reviewer 2

2.1 Novelty, and whether re-crawling or live reading removes the problem

Concern. State what is new beyond showing that a snapshot goes stale, and whether re-crawling or live reading removes the problem; the data show 0 mismatches under live reading on Gitea.

Response. Live reading removes the measured false proofs on both products: 0 mismatches under the live reading against 13 stale false proofs on Gitea and 16 on GitLab, and a recrawl with the same credential repairs 29 of 29. We removed the unconditional claim that more crawling is ineffective, and the text now says when a recrawl suffices—when the model can be refreshed with the same credential before adjudication—and when it does not: a grant the surface cannot enumerate at all is invisible to any recrawl, which is the out-of-band channel where the residual false proofs sit. What is new is therefore not the observation that a snapshot goes stale, but the measurement of when staleness produces a *false proof* rather than a miss, the separation of that mechanism from an unsound oracle, and the fact that the residual survives the strongest tester-side bookkeeping.

Action. Section V-A (Table 12, p. 14); Section VI-A; the related-work claim restated in Section IX.

2.2 Scope of claims: two relations of one family

Concern. Only 2 of 76 relations were studied and only 2 of 11 authorization relations fired on Gitea; the title and abstract should be narrowed or the evaluation extended.

Response. The title, the running head, the PDF title and the abstract are narrowed to the two reachability-gated relations in MST-wi, and the abstract states the family restriction and the native trigger range: of the 11 catalog authorization relations, 10 were executed against Gitea and only the reachability-gated ones alarm. We did not extend the evaluation to other relation families; Section VIII names this as the paper's single most important limitation and says what raising it would require—relations gated otherwise, which is a different experiment.

Action. Title, running head, abstract, Section VIII.

2.3 Proposition 2

Concern. It is close to true by definition; separate what is empirical from what is definitional.

Response. Done as described under comment 5: the algebraic absorption is a lemma and is labelled definitional, the claim that these two relations fall in its domain is a separate proposition with an empirical premise that is checked against the records, and the truth table makes the functional representation explicit.

Action. Section VI-B, Table 16 (p. 17).

2.4 Detection side: no real system was vulnerable

Concern. Either add a real vulnerable case or remove the claim that the metamorphic step adds nothing for detection.

Response. Both halves were done. A real vulnerable/fixed pair was added (Section V-D, Table 15, p. 15): a publicly fixed Gitea defect, one true positive on the vulnerable build, no claim on the fixed build, twelve control checks passing. And the claim was narrowed: the manuscript no longer says the metamorphic step adds nothing *for detection*; what it supports is that the step adds no adjudication information where a readable, sound oracle exists (Lemma 1 and Proposition 2), and that the relations' recall on real defects is unevaluated—which Section X states as the boundary of the evidence.

Action. Section V-D and Table 15; Section VI-B; Section X.

2.5 The 84.62% figure

Concern. It assumes two inputs per case and is only illustrative; keep it out of the abstract and do not present it as quantitative support.

Response. The figure is absent from the abstract and has now been removed from the body as well; it appears nowhere in the manuscript and is not used as quantitative support for anything. The denominator argument is made in parameterized form instead (Section IV-E).

Action. Section IV-E; verified absent from the compiled manuscript.

2.6 Constructed ground truth and the reported rates

Concern. Discuss how far the false-proof rates reflect the scenario design, such as the per-cell multipliers, rather than the technique.

Response. The manuscript states in several places that the channel proportions are topology design parameters and not field prevalence, that the eight transitions are a designed set and not a sample of deployment changes, and that the out-of-band share quoted on the real system is a property of the constructed topology. The design dependence is also made inspectable rather than asserted: Table 5 gives the per-block record counts behind the 312/264 partition, Table 19 gives the per-cell census, and the matched-pair attribution (Table 2) shows exactly which cells carry each effect—the false-proof saving falls on one scenario, the legitimate share created through the framework's own facility, while the out-of-band scenario agrees with the reference. The reported rates are therefore presented as properties of the construction, not as deployment rates.

Action. Sections V-B, VI-A and VIII; Tables 2, 5 and 19.

What remains. We did not recompute the constructed rates under equal per-cell weights in parallel with the original weights; the sensitivity to the weighting is described qualitatively rather than tabulated.

3 Answers to the additional questions

Reviewer 1. (1) Contribution to the body of knowledge: yes, in the narrowed form stated in Section IX and Table 18—a predicate-level account of what the two relations determine, a channel decomposition of the false proofs, and a resolution argument about a pooled specificity. (2) Technically sound: the algebraic part is separated from the empirical part and the empirical part is checked against records by script; the unsupported parts are marked as limitations. (3) Comprehensive: the design, the ablation, the real-system studies and the reproducibility package are all reported, with the gaps named. (4) and (5) References: applicable and sufficient; none is inappropriate.

Reviewer 2. (1) Contribution: partly, and we have narrowed the claim accordingly rather than defending the broader reading. (2) Technically sound: the method is transparent and the artefacts are public; the evidence remains narrow and is labelled as such. (3) Comprehensive: the related work is retained and the positioning claim is now falsifiable. (4) and (5) References: applicable and sufficient; none is inappropriate.

4 What this revision does not claim

For the Associate Editor's convenience, the eight items that are met only in part, with the single missing element in each:

Item	What was done	What is still missing
R1.2	Truth table over (Y,M,O) with E split; A, D and H defined; the engine-versus-policy split reported	The E=unknown contract is stated across several sections rather than collected in one place
R1.4	Pre-registered protocol bound by digest; primary-vantage gate checked before the run; amendment timing disclosed	A mechanical, implementation- independent test of independence from the tested mechanism
R1.8	Per-channel four-cell matrix for Gitea and GitLab; the two failure mechanisms separated	The same matrix for Gogs, whose surface exposes no equivalent decomposition
R1.9 / R2.4	A real vulnerable/fixed pair through the adjudication gate; the seeded control bounded to one fault pattern	Per-defect recall and false-positive rates across several independent defect classes
R1.12	Fixed-weight derivation; arbitrary class-share bound	A derivation for detector-endogenous input counts
R1.14	Pinned environment, commands, seeds, raw outputs, regeneration scripts, clean-copy verification	The failure and missing-observation rules stated in the body rather than only in the protocol file and scripts
R2.6	Design dependence stated; per-block and per-cell counts inspectable	Rates recomputed under equal per-cell weights alongside the original weights

We would rather submit with these eight named than with a text that implies they are closed. Each is visible in the manuscript as a stated boundary, in Section VIII or at the point of use.

5 Files in this resubmission

1. This response to reviewers.
2. The revised manuscript with all changes highlighted.
3. A clean copy of the revised manuscript, as PDF and as the LaTeX source package (90 files), which compiles in an empty directory and reproduces every table and figure from the raw records.